HRI AGRRA Belize — Species-Level EDA

Individual fish family biomass distributions, trends, and spatial patterns

2026-03-15

Overview

This document examines individual fish family biomass patterns across the Belize Barrier Reef, moving beyond the grouped commercial/herbivorous totals used in the initial analysis. Seven fish families are examined: 5 commercial (Snappers, Groupers, Jacks, Barracuda, Grunts) and 2 herbivorous (Surgeonfish, Parrotfish).

```
# Observations with fish data
fish_data <- full %>% filter(!is.na(tCommBiomass) | !is.na(tHerbiBiomass))

fish_data %>%
  summarise(
    `Total obs` = n(),
    `With commercial` = sum(!is.na(tCommBiomass)),
    `With herbivorous` = sum(!is.na(tHerbiBiomass)),
    `Years` = paste(unique(YEAR), collapse = ", ")
  ) %>%
  knitr::kable()
```

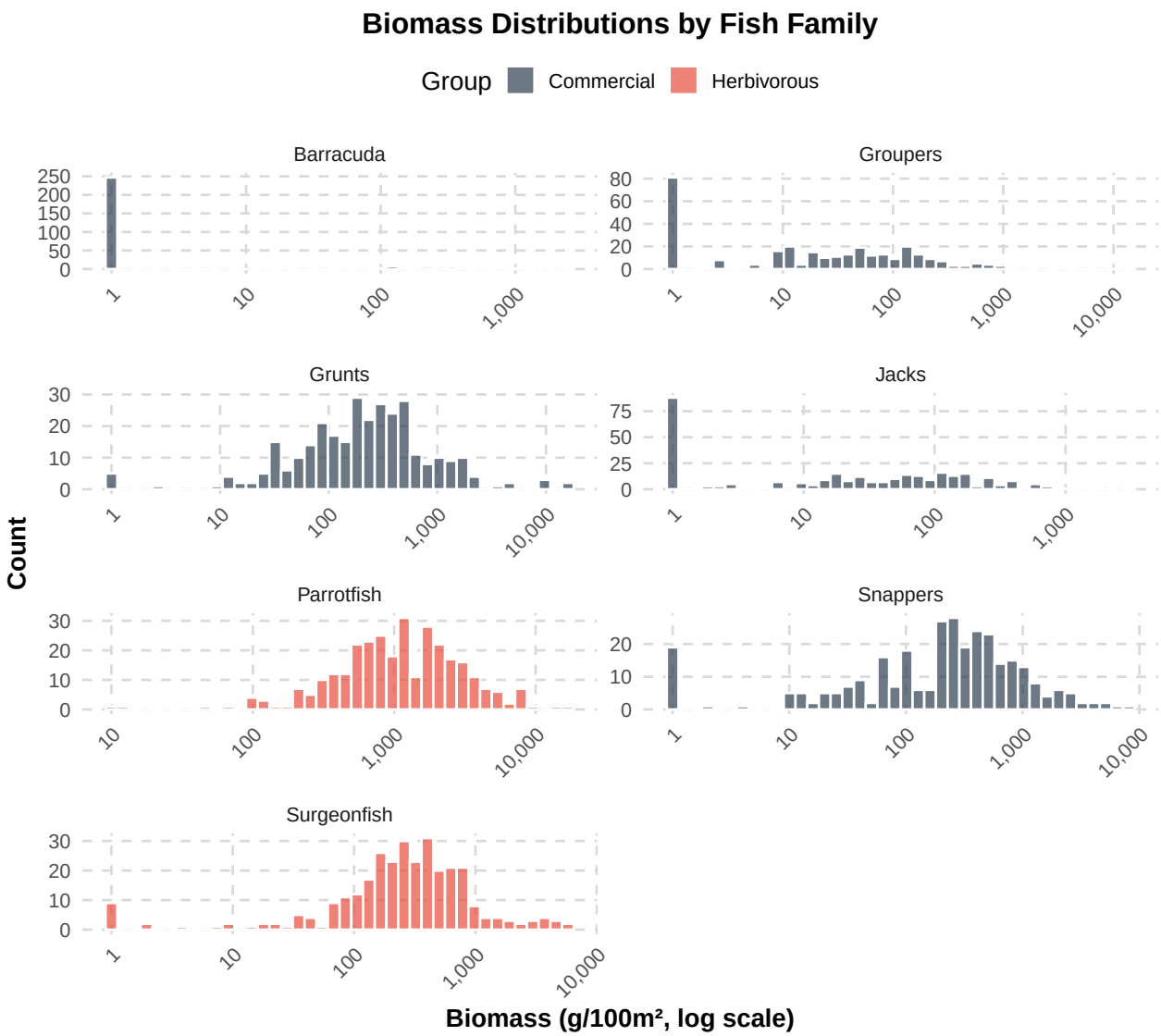
Total obs	With commercial	With herbivorous	Years
308	308	308	2018, 2023, 2021

Species Distributions

Biomass distributions by family

```
fish_long <- full %>%
  select(YEAR, all_of(species_cols)) %>%
  pivot_longer(-YEAR, names_to = "species", values_to = "biomass") %>%
  filter(!is.na(biomass)) %>%
  mutate(
    common = species_labels[species],
    group  = species_info$group[match(species, species_info$code)]
  )
```

```r
ggplot(fish_long, aes(x = biomass + 1, fill = group)) +
  geom_histogram(bins = 40, alpha = 0.7, color = "white") +
  scale_x_log10(labels = scales::comma) +
  scale_fill_manual(values = group_colors) +
  facet_wrap(~common, scales = "free", ncol = 2) +
  labs(x = "Biomass (g/100m², log scale)", y = "Count", fill = "Group",
       title = "Biomass Distributions by Fish Family") +
  theme_hri()
```



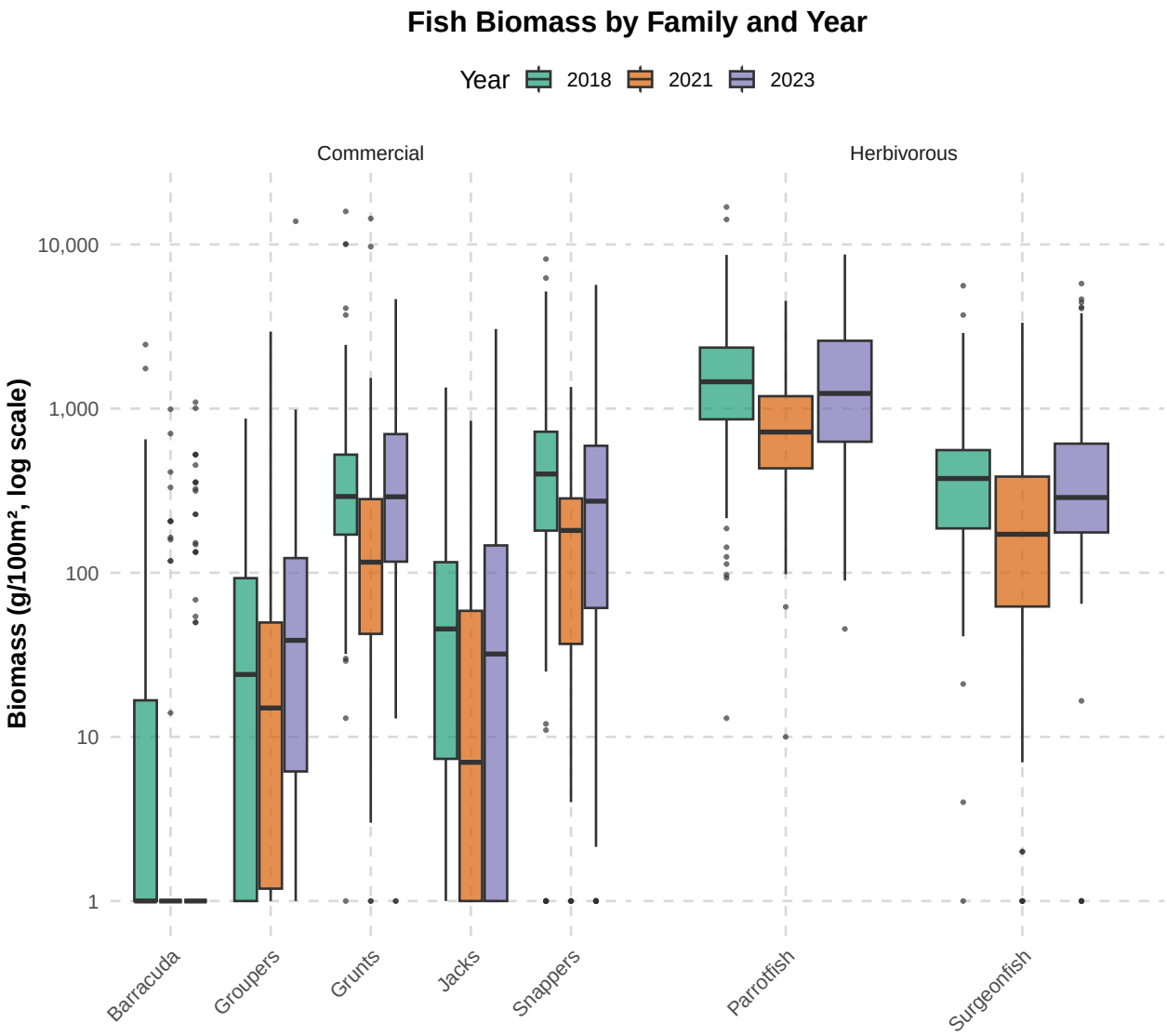
Summary statistics

```r
fish_long %>%
  group_by(common, group) %>%
  summarise(
    n       = n(),
    mean    = round(mean(biomass, na.rm = TRUE), 1),
    median  = round(median(biomass, na.rm = TRUE), 1),
    sd      = round(sd(biomass, na.rm = TRUE), 1),
    min     = round(min(biomass, na.rm = TRUE), 1),
    max     = round(max(biomass, na.rm = TRUE), 1),
    pct_zero = round(100 * mean(biomass == 0, na.rm = TRUE), 1),
    .groups = "drop"
  ) %>%
  arrange(group, desc(mean)) %>%
  knitr::kable(col.names = c("Family", "Group", "n", "Mean", "Median", "SD",
                             "Min", "Max", "% Zero"))
```

Family	Group	n	Mean	Median	SD	Min	Max	% Zero
Grunts	Commercial	308	598.4	218.0	1612.9	0	15890.0	1.6
Snappers	Commercial	308	527.2	251.2	920.3	0	8153.0	6.2
Groupers	Commercial	308	135.6	23.0	814.6	0	13835.7	26.3
Jacks	Commercial	308	102.8	24.2	247.5	0	3053.4	28.6
Barracuda	Commercial	308	64.7	0.0	226.0	0	2453.0	80.2
Parrotfish	Herbivorous	308	1724.5	1144.6	1975.7	9	16914.0	0.0
Surgeonfish	Herbivorous	308	513.4	272.8	815.0	0	5777.3	2.9

Boxplots by family and year

```r
ggplot(fish_long, aes(x = common, y = biomass + 1, fill = YEAR)) +
  geom_boxplot(alpha = 0.7, outlier.size = 0.8) +
  scale_y_log10(labels = scales::comma) +
  scale_fill_brewer(palette = "Dark2") +
  facet_wrap(~group, scales = "free_x") +
  labs(x = NULL, y = "Biomass (g/100m²", log scale)", fill = "Year",
       title = "Fish Biomass by Family and Year") +
  theme_hri()
```

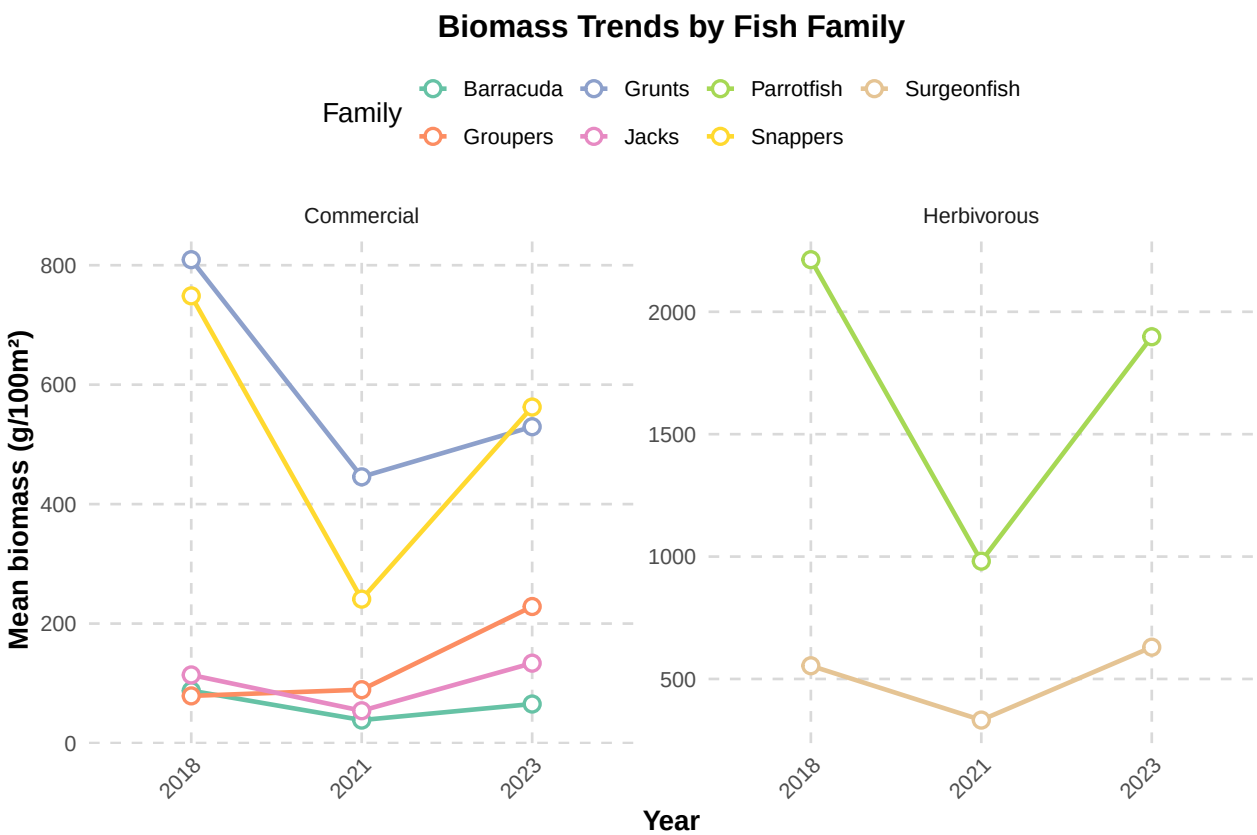


Temporal Trends

Mean biomass by family over time

```
trends <- fish_long %>%
  group_by(YEAR, common, group) %>%
  summarise(mean_bio = mean(biomass, na.rm = TRUE), .groups = "drop")

ggplot(trends, aes(x = YEAR, y = mean_bio, group = common, color = common)) +
  geom_line(linewidth = 1.1) +
```

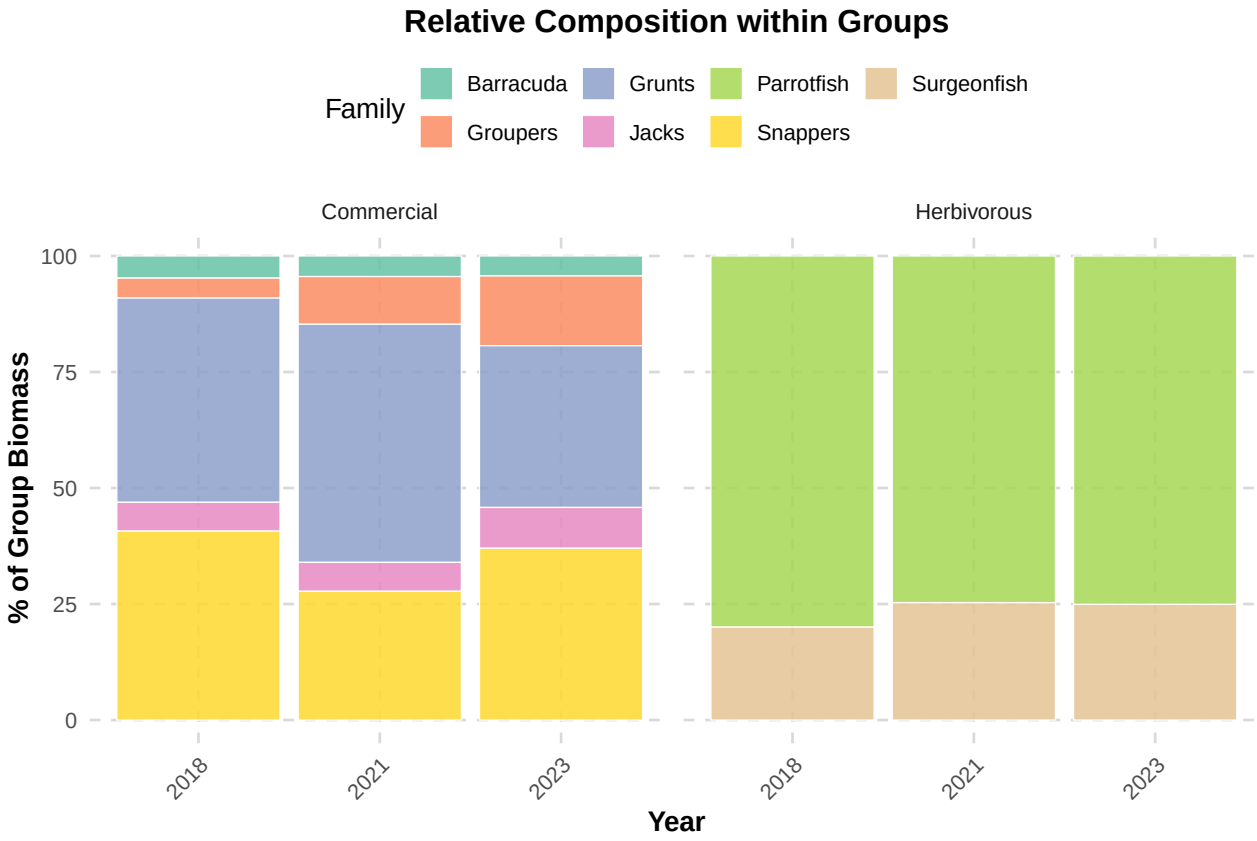
```
  geom_point(size = 3, shape = 21, fill = "white", stroke = 1.2) +
  facet_wrap(~group, scales = "free_y") +
  scale_color_brewer(palette = "Set2") +
  labs(x = "Year", y = "Mean biomass (g/100m²)", color = "Family",
       title = "Biomass Trends by Fish Family") +
  theme_hri()
```



Relative composition over time

```
comp <- fish_long %>%
  group_by(YEAR, group, common) %>%
  summarise(total = sum(biomass, na.rm = TRUE), .groups = "drop") %>%
  group_by(YEAR, group) %>%
  mutate(pct = 100 * total / sum(total))

ggplot(comp, aes(x = YEAR, y = pct, fill = common)) +
  geom_col(alpha = 0.85, color = "white", linewidth = 0.3) +
  facet_wrap(~group) +
  scale_fill_brewer(palette = "Set2") +
```

```
  labs(x = "Year", y = "% of Group Biomass", fill = "Family",
       title = "Relative Composition within Groups") +
  theme_hri()
```

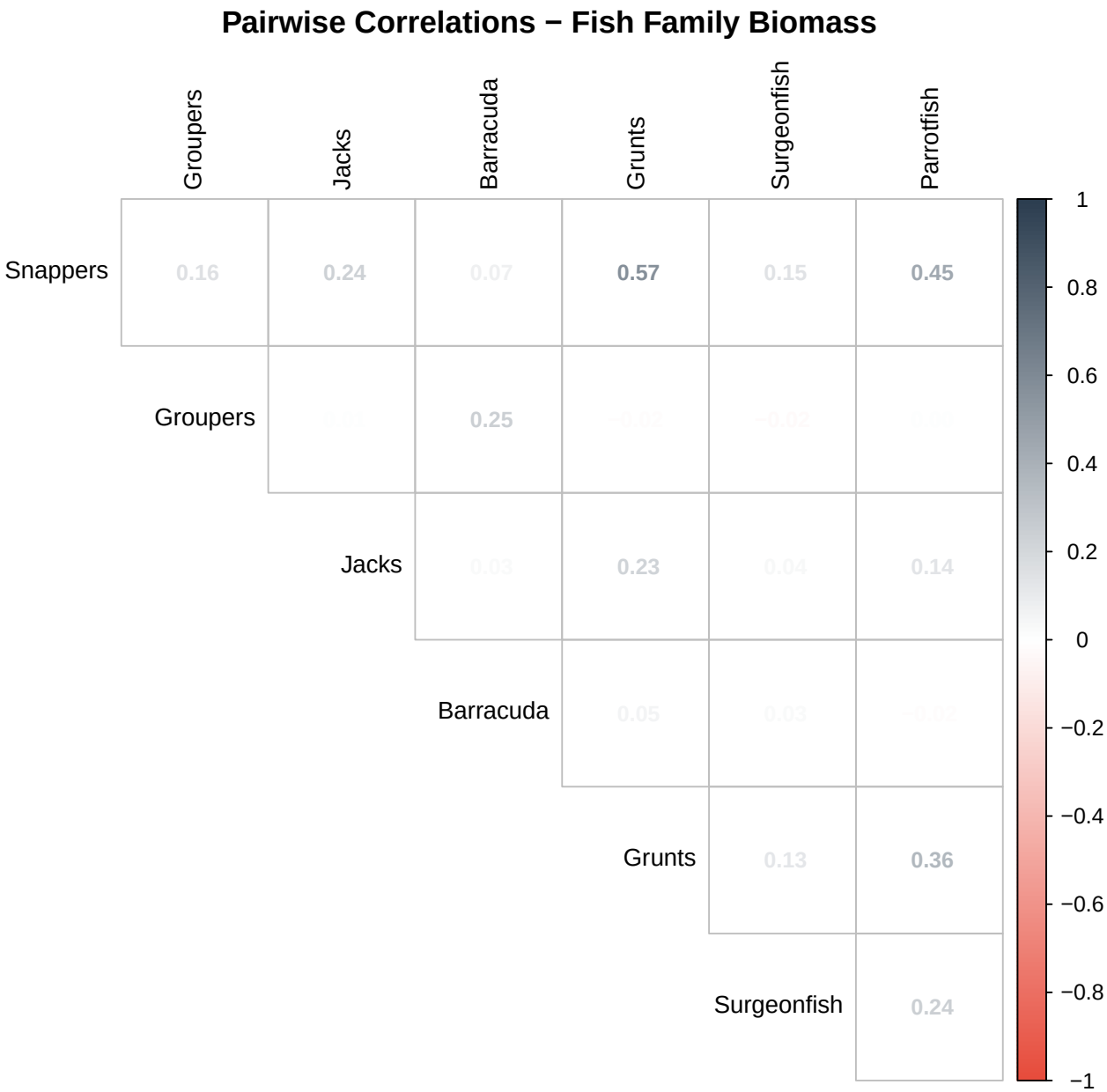


Species Correlations

Pairwise correlations among all 7 families

```
fish_wide <- full %>%
  select(all_of(species_cols)) %>%
  rename_with(~species_labels[.x]) %>%
  na.omit()

cor_mat <- cor(fish_wide, use = "pairwise.complete.obs")

corrplot(cor_mat, method = "number", type = "upper", diag = FALSE,
         number.cex = 0.85, tl.cex = 0.9, tl.col = "black",
         col = colorRampPalette(c("#E74C3C", "white", "#2C3E50"))(200),
         title = "Pairwise Correlations - Fish Family Biomass",
```

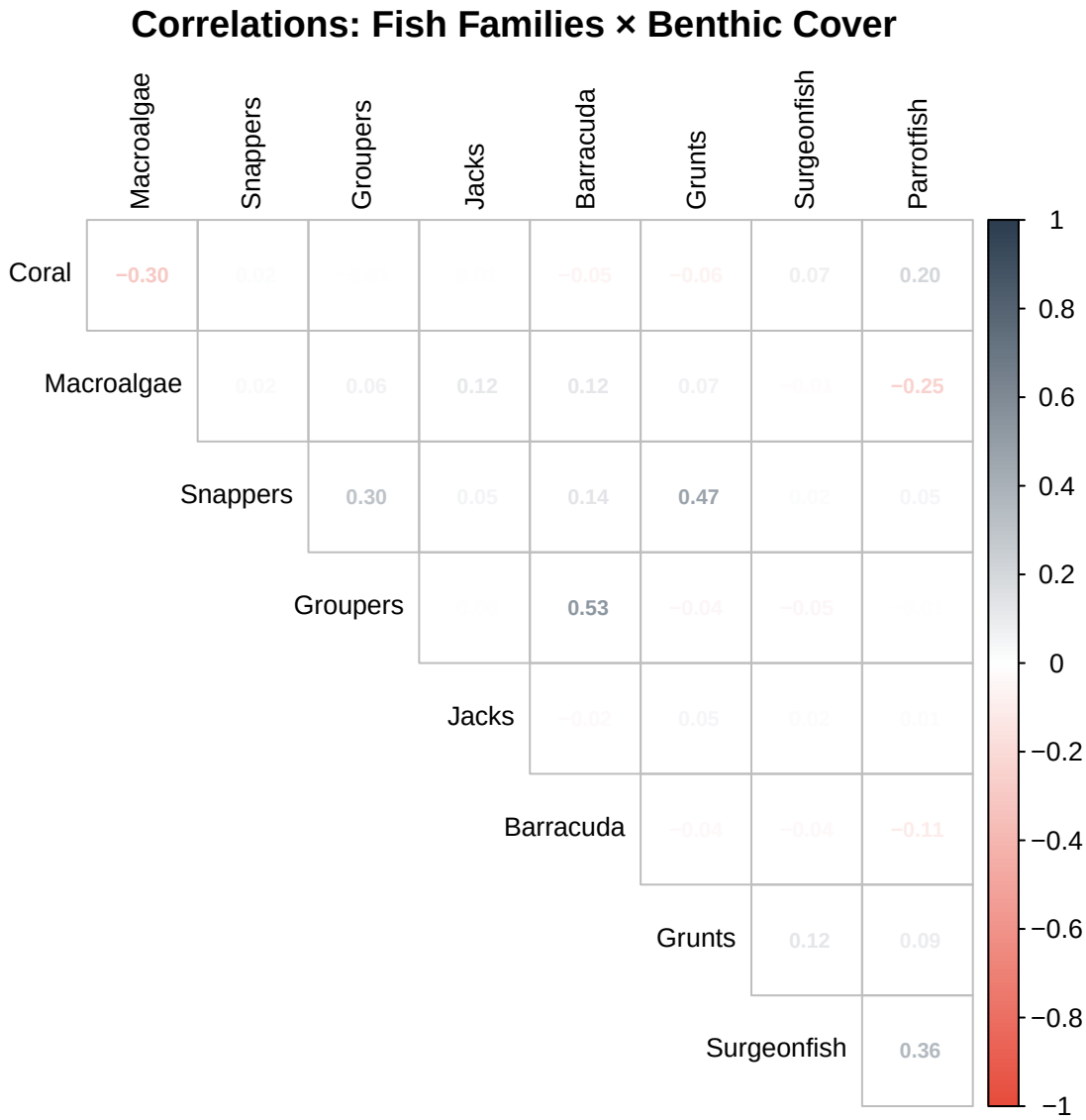
```
        mar = c(0, 0, 2, 0))
```



Correlations with benthic cover

```
benthic_fish <- full %>%
  select(tCORALavg, tFMAavg, all_of(species_cols)) %>%
```

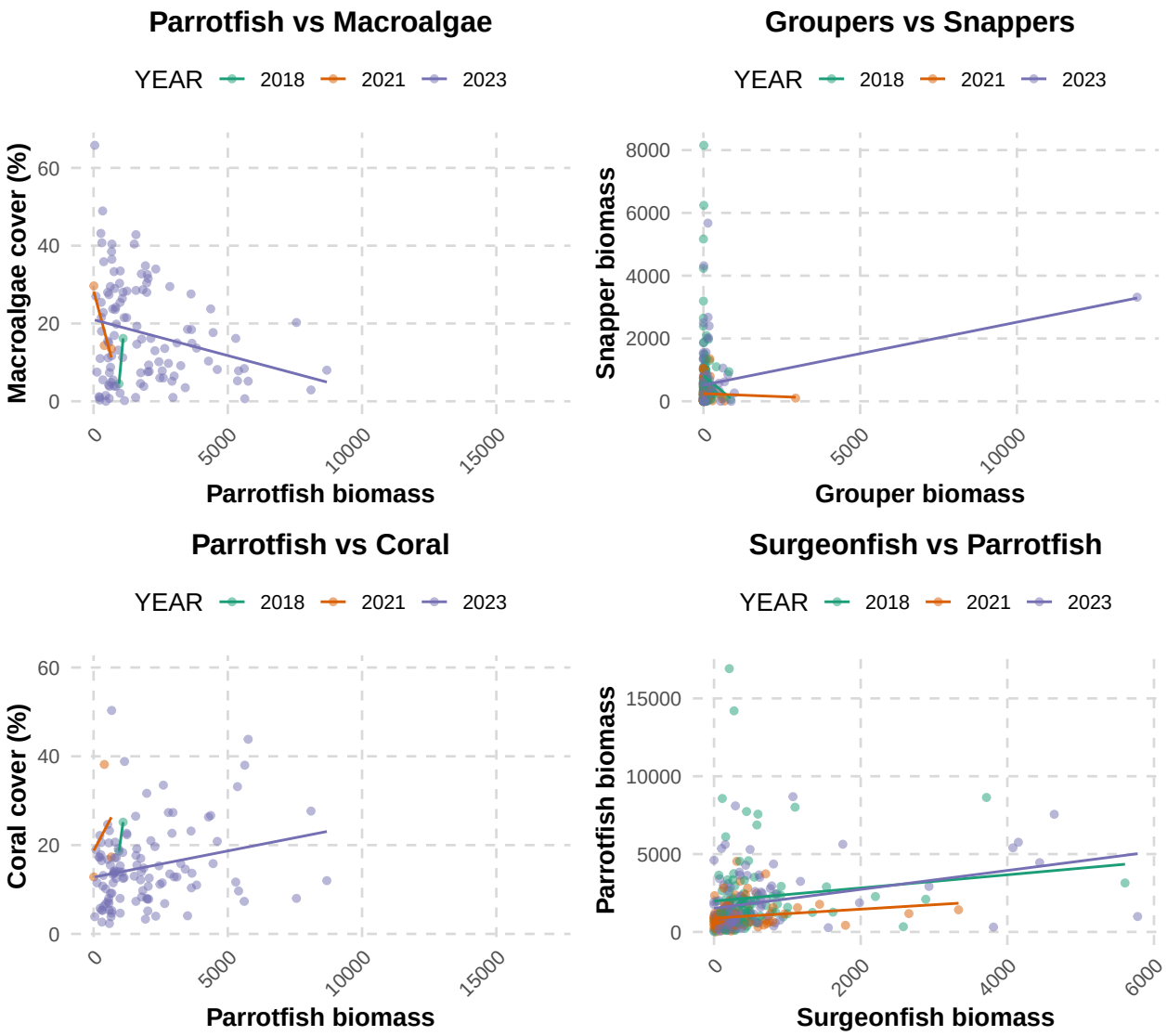
```r
  rename(Coral = tCORALavg, Macroalgae = tFMAavg) %>%
  rename_with(~ifelse(.x %in% names(species_labels), species_labels[.x], .x)) %>%
  na.omit()

cor_bf <- cor(benthic_fish)

corrplot(cor_bf, method = "number", type = "upper", diag = FALSE,
         number.cex = 0.7, tl.cex = 0.85, tl.col = "black",
         col = colorRampPalette(c("#E74C3C", "white", "#2C3E50"))(200),
         title = "Correlations: Fish Families × Benthic Cover",
         mar = c(0, 0, 2, 0))
```



Key scatterplots

```
p_pairs <- list()

# Parrotfish vs macroalgae (herbivory hypothesis)
p_pairs[[1]] <- ggplot(full, aes(x = tSCARavg, y = tFMAavg, color = YEAR)) +
  geom_point(alpha = 0.5, size = 1.5) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 0.7) +
  scale_color_brewer(palette = "Dark2") +
  labs(x = "Parrotfish biomass", y = "Macroalgae cover (%)",
```

```r
       title = "Parrotfish vs Macroalgae") +
  theme_hri()

# Groupers vs snappers (competition/predation)
p_pairs[[2]] <- ggplot(full, aes(x = tSERRavg, y = tLUTJavg, color = YEAR)) +
  geom_point(alpha = 0.5, size = 1.5) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 0.7) +
  scale_color_brewer(palette = "Dark2") +
  labs(x = "Grouper biomass", y = "Snapper biomass",
       title = "Groupers vs Snappers") +
  theme_hri()

# Parrotfish vs coral (bioerosion/facilitation)
p_pairs[[3]] <- ggplot(full, aes(x = tSCARavg, y = tCORALavg, color = YEAR)) +
  geom_point(alpha = 0.5, size = 1.5) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 0.7) +
  scale_color_brewer(palette = "Dark2") +
  labs(x = "Parrotfish biomass", y = "Coral cover (%)",
       title = "Parrotfish vs Coral") +
  theme_hri()

# Surgeonfish vs parrotfish (herbivore competition)
p_pairs[[4]] <- ggplot(full, aes(x = tACANavg, y = tSCARavg, color = YEAR)) +
  geom_point(alpha = 0.5, size = 1.5) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 0.7) +
  scale_color_brewer(palette = "Dark2") +
  labs(x = "Surgeonfish biomass", y = "Parrotfish biomass",
       title = "Surgeonfish vs Parrotfish") +
  theme_hri()

gridExtra::grid.arrange(grobs = p_pairs, ncol = 2)
```



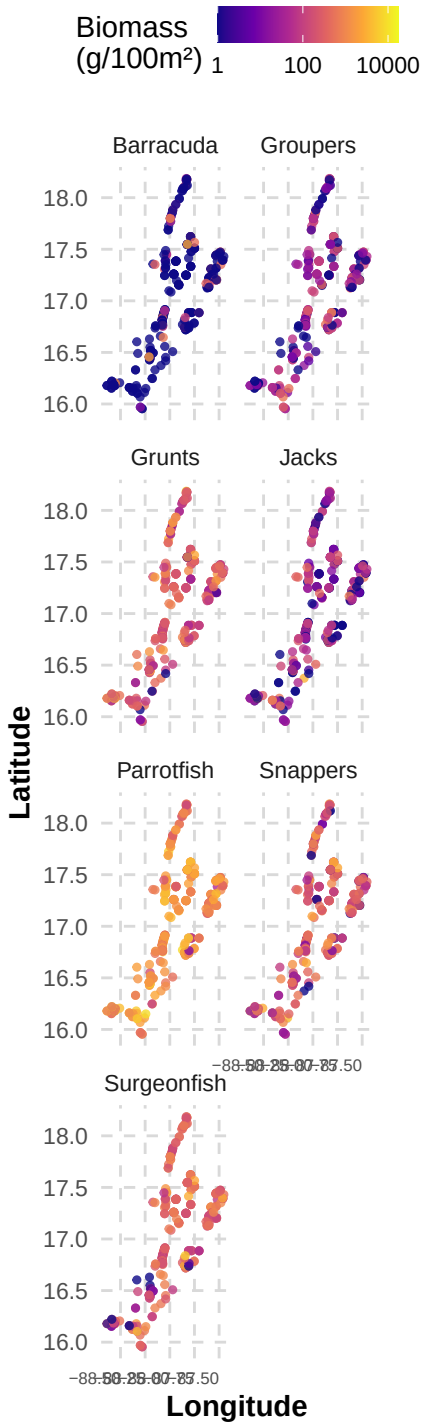
Spatial Patterns

Individual species spatial distributions

```
# Site-level averages for mapping
site_avg <- full %>%
  group_by(Name, Latitude, Longitude) %>%
  summarise(across(all_of(species_cols), ~mean(.x, na.rm = TRUE)),
            .groups = "drop")
```

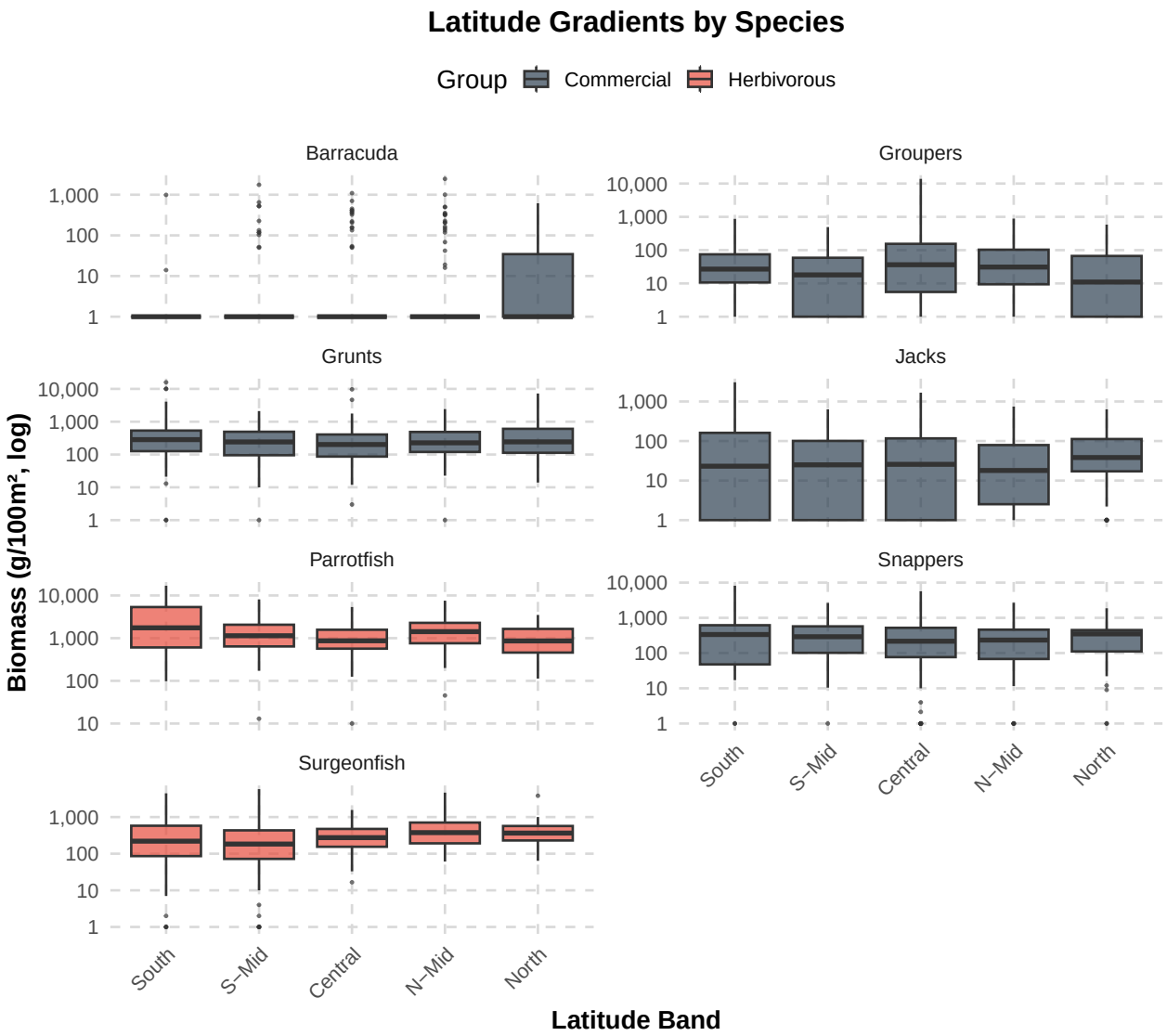
```r
site_long <- site_avg %>%
  pivot_longer(all_of(species_cols), names_to = "species", values_to = "biomass") %>%
  mutate(common = species_labels[species],
         group  = species_info$group[match(species, species_info$code)])

ggplot(site_long, aes(x = Longitude, y = Latitude, color = biomass + 1)) +
  geom_point(size = 1.2, alpha = 0.8) +
  scale_color_viridis_c(option = "C", trans = "log10", name = "Biomass\n(g/100m²)") +
  facet_wrap(~common, ncol = 2) +
  coord_quickmap() +
  labs(title = "Spatial Distribution - Fish Family Biomass") +
  theme_hri() +
  theme(axis.text.x = element_text(angle = 0, size = 8))
```

Spatial Distribution − Fish Family Biomass



Latitude gradients by species

```
site_long %>%
  mutate(lat_band = cut(Latitude, breaks = 5, labels = c("South", "S-Mid",
                                                          "Central", "N-Mid", "North"))) %>%
  ggplot(aes(x = lat_band, y = biomass + 1, fill = group)) +
  geom_boxplot(alpha = 0.7, outlier.size = 0.5) +
  scale_y_log10(labels = scales::comma) +
  scale_fill_manual(values = group_colors) +
  facet_wrap(~common, scales = "free_y", ncol = 2) +
  labs(x = "Latitude Band", y = "Biomass (g/100m², log)", fill = "Group",
       title = "Latitude Gradients by Species") +
  theme_hri()
```

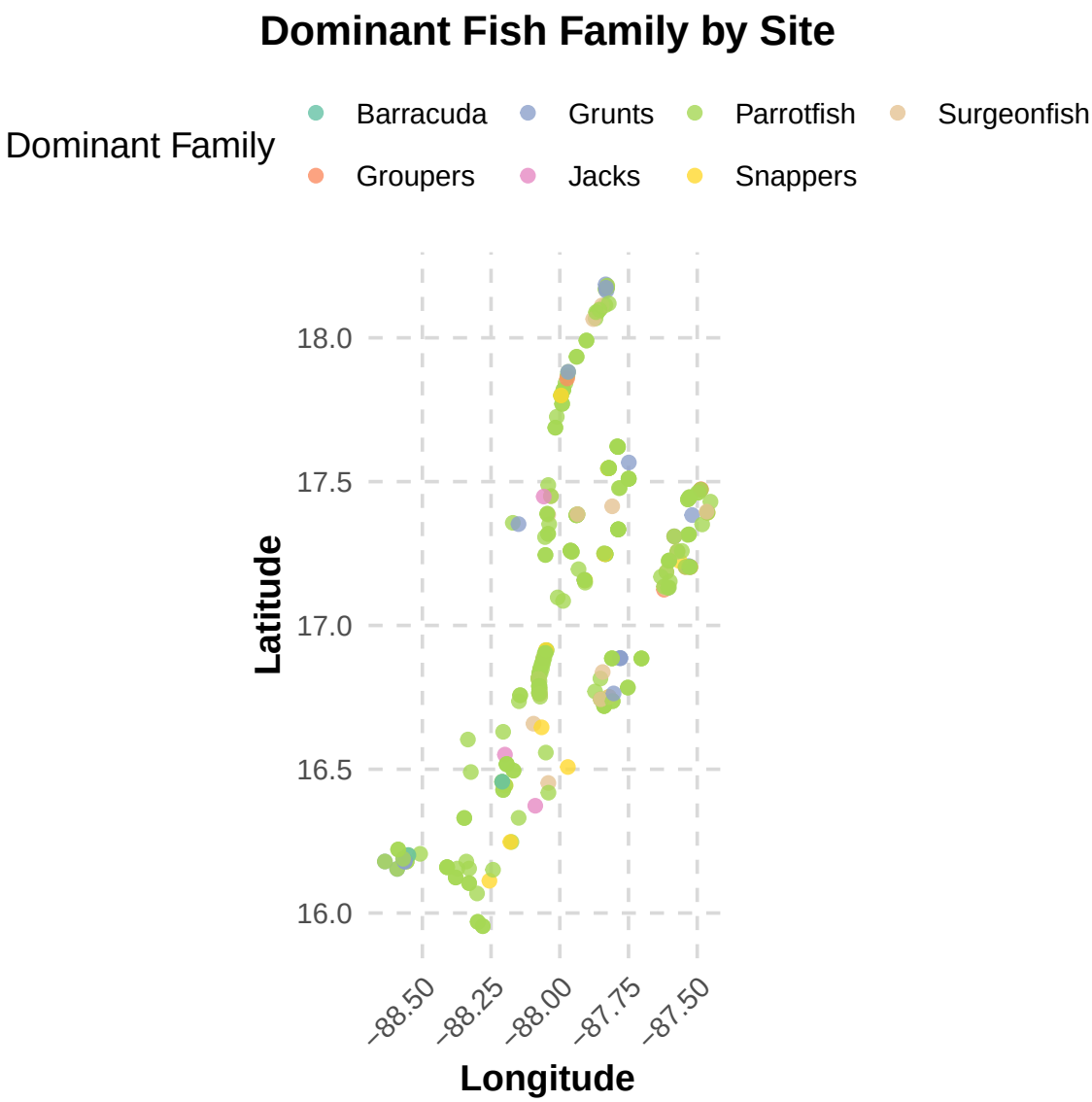


Dominance and Evenness

Which families dominate at each site?

```
dominant <- site_avg %>%
  pivot_longer(all_of(species_cols), names_to = "species", values_to = "biomass") %>%
  group_by(Name, Latitude, Longitude) %>%
  slice_max(biomass, n = 1, with_ties = FALSE) %>%
  ungroup() %>%
  mutate(common = species_labels[species])
```

```r
ggplot(dominant, aes(x = Longitude, y = Latitude, color = common)) +
  geom_point(size = 2, alpha = 0.8) +
  scale_color_brewer(palette = "Set2") +
  coord_quickmap() +
  labs(color = "Dominant Family",
       title = "Dominant Fish Family by Site") +
  theme_hri()
```



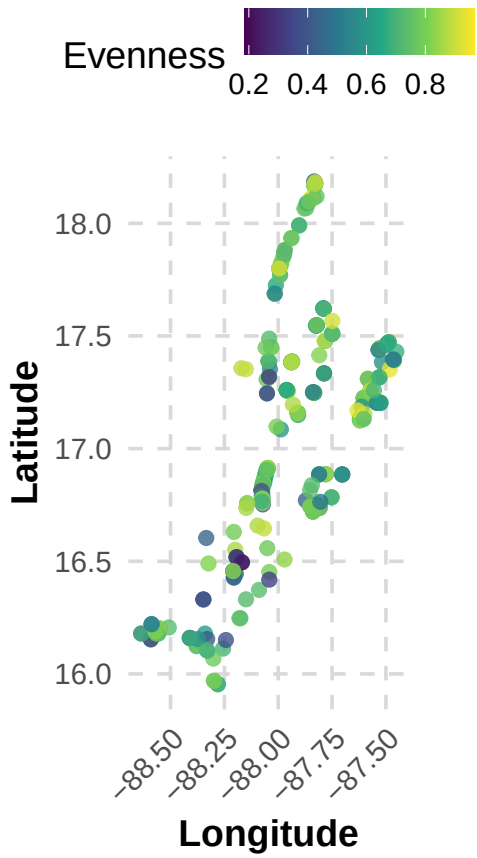

```r
dominant %>%
  count(common) %>%
```

```r
  mutate(pct = round(100 * n / sum(n), 1)) %>%
  arrange(desc(n)) %>%
  knitr::kable(col.names = c("Dominant Family", "Sites", "% of Sites"))
```

Dominant Family	Sites	% of Sites
Parrotfish	220	75.6
Surgeonfish	26	8.9
Grunts	20	6.9
Snappers	14	4.8
Groupers	6	2.1
Jacks	3	1.0
Barracuda	2	0.7

Simpson's evenness across families

```r
evenness <- site_avg %>%
  rowwise() %>%
  mutate(
    total = sum(c_across(all_of(species_cols)), na.rm = TRUE),
    simpson_d = 1 - sum((c_across(all_of(species_cols)) / total)^2, na.rm = TRUE),
    evenness = simpson_d / (1 - 1/7)  # normalized by max possible
  ) %>%
  ungroup()

ggplot(evenness, aes(x = Longitude, y = Latitude, color = evenness)) +
  geom_point(size = 2, alpha = 0.8) +
  scale_color_viridis_c(option = "D", name = "Evenness") +
  coord_quickmap() +
  labs(title = "Fish Community Evenness (Simpson's)") +
  theme_hri()
```

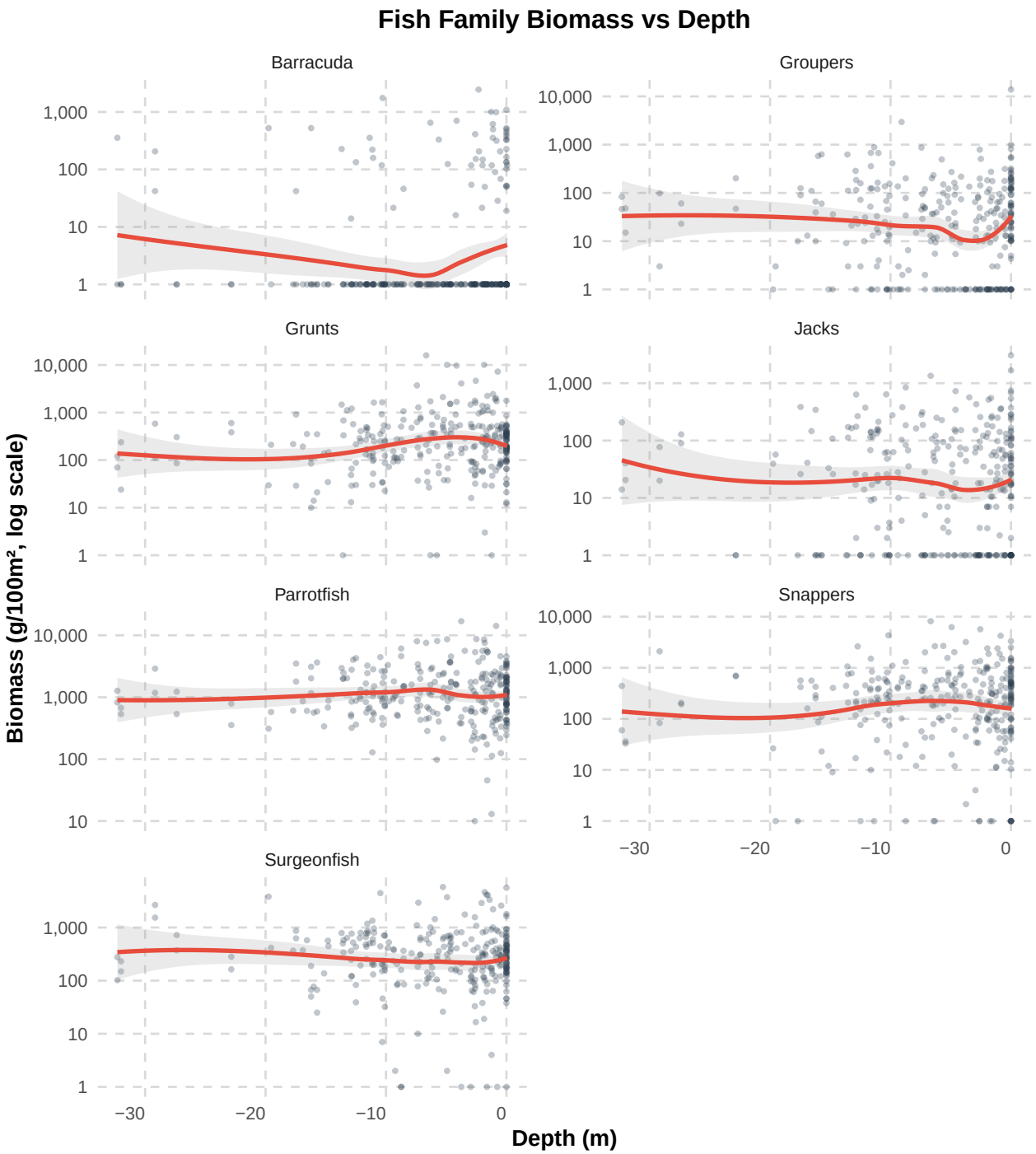


Predictor Relationships

Species biomass vs bathymetric predictors

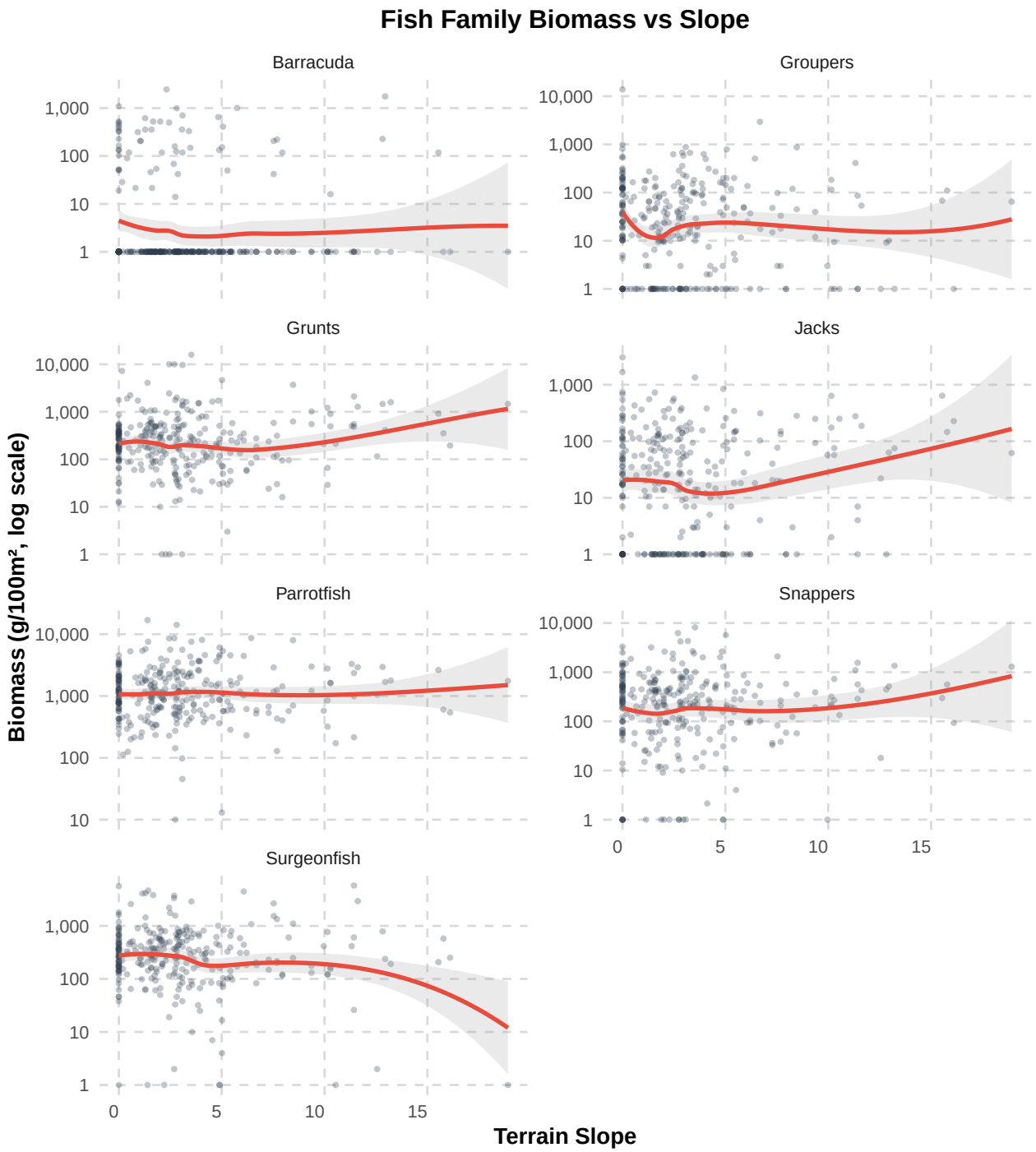
```
hri <- read.csv(here("data", "external", "HRI_all_variables.csv"))

# Merge individual species with predictors on coordinates
site_pred <- site_avg %>%
  inner_join(
    hri %>% rename(Longitude = Longitude_, Latitude = Latitude_y),
    by = c("Longitude", "Latitude")
  )

# Depth relationships
depth_long <- site_pred %>%
  select(depth, all_of(species_cols)) %>%
```

```r
  pivot_longer(-depth, names_to = "species", values_to = "biomass") %>%
  mutate(common = species_labels[species])

ggplot(depth_long, aes(x = depth, y = biomass + 1)) +
  geom_point(alpha = 0.3, size = 1, color = "#2C3E50") +
  geom_smooth(method = "loess", color = "#E74C3C", se = TRUE, alpha = 0.2) +
  scale_y_log10(labels = scales::comma) +
  facet_wrap(~common, scales = "free_y", ncol = 2) +
  labs(x = "Depth (m)", y = "Biomass (g/100m², log scale)",
       title = "Fish Family Biomass vs Depth") +
  theme_hri() +
  theme(axis.text.x = element_text(angle = 0))
```


Fish Family Biomass vs Depth
Barracuda
Groupers
Grunts
Jacks
Parrotfish
Snappers
Surgeonfish
Biomass (g/100m², log scale)
Depth (m)
1
10
100
1,000
10,000
−30
−20
−10
0

Slope relationships

```
slope_long <- site_pred %>%
  select(slope, all_of(species_cols)) %>%
  pivot_longer(-slope, names_to = "species", values_to = "biomass") %>%
  mutate(common = species_labels[species])

ggplot(slope_long, aes(x = slope, y = biomass + 1)) +
  geom_point(alpha = 0.3, size = 1, color = "#2C3E50") +
  geom_smooth(method = "loess", color = "#E74C3C", se = TRUE, alpha = 0.2) +
  scale_y_log10(labels = scales::comma) +
  facet_wrap(~common, scales = "free_y", ncol = 2) +
  labs(x = "Terrain Slope", y = "Biomass (g/100m², log scale)",
       title = "Fish Family Biomass vs Slope") +
  theme_hri() +
  theme(axis.text.x = element_text(angle = 0))
```


Fish Family Biomass vs Slope
Barracuda
Groupers
Grunts
Jacks
Parrotfish
Snappers
Surgeonfish
Biomass (g/100m², log scale)
Terrain Slope
1,000
100
10
1
10,000
0
5
10
15

Summary

```r
fish_long %>%
  group_by(common, group) %>%
  summarise(
    `Mean biomass` = round(mean(biomass, na.rm = TRUE), 1),
    `Dominant at n sites` = sum(dominant$common == first(common)),
    .groups = "drop"
  ) %>%
  arrange(group, desc(`Mean biomass`)) %>%
  knitr::kable()
```

common	group	Mean biomass	Dominant at n sites
Grunts	Commercial	598.4	20
Snappers	Commercial	527.2	14
Groupers	Commercial	135.6	6
Jacks	Commercial	102.8	3
Barracuda	Commercial	64.7	2
Parrotfish	Herbivorous	1724.5	220
Surgeonfish	Herbivorous	513.4	26

Key observations:

- Individual species show distinct spatial patterns not captured by grouped totals
- Parrotfish and Grunts tend to dominate biomass at most sites
- Species-specific depth preferences suggest niche partitioning along the depth gradient
- Pairwise correlations among families reveal both positive (co-occurrence) and negative (competitive exclusion) relationships
- Community evenness varies spatially, with potential north-south gradients along the barrier reef